JUBORAJ AHMED

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Education

BRAC University *Dhaka, Bangladesh*
Bachelor of Science in Computer Science 2026

Overall GPA: 2.98/4.00 **Final 60 Credits GPA:** 3.00/4.00

Research Interests

Trustworthy Medical AI • Medical Image Analysis • Vision-Language Models • Medical AI Foundation Models • Multimodal Learning • Robustness and Reliability of Clinical AI

Research Experience

Trustworthy Vision-Language Models for Medical Image Understanding
Ongoing Research Collaboration *2026–Present*

- Investigating the reliability of medical vision-language models when image-grounded clinical evidence conflicts with misleading textual instructions, authority cues, or social pressure.
- Designing controlled experiments to examine **prompt-induced diagnostic capitulation**, assessing whether models abandon evidence-supported medical interpretations under conflicting contextual information.
- Studying diagnostic consistency, multimodal grounding, and failure patterns under matched prompting conditions to better understand robustness in medical image interpretation.

Directional Constraints for Patient-Specific Alert Thresholds in Early Sepsis Monitoring: An Empirical Bayes Analysis
Completed Manuscript; preparing for journal submission *2026*

- Investigated a failure mode of personalized clinical alerting in which early disease-related risk is misinterpreted as stable patient heterogeneity, potentially elevating alert thresholds and suppressing early warnings.
- Developed and evaluated **Directionally Constrained Empirical Bayes (DC-EB)** threshold adaptation, restricting upward personalization while retaining patient-specific downward adjustment.
- Evaluated thresholding strategies on the **PhysioNet/CinC 2019 sepsis dataset** comprising 40,336 ICU patients, using 8-, 12-, and 24-hour adaptation histories, patient-level data partitions, and **2,000 paired bootstrap replicates**.
- Found that directional constraints mitigated sensitivity and normalized clinical-utility losses associated with unrestricted personalization, although superiority over the optimized Global threshold was not conclusively established.

JASAL: A Remote Patient Health Monitoring System Driven by Federated Learning
Undergraduate Thesis, BRAC University; supervised by M. Mostakim *2025–2026*

- Co-developed a federated learning framework for privacy-aware physiological anomaly detection using **32,796 records across 10 non-IID client partitions**, modeling decentralized healthcare environments.
- Designed adaptive patient profiling and reconstruction-error-based feature attribution to support personalized anomaly detection and interpretable physiological monitoring.
- Integrated differential privacy, secure aggregation mechanisms, and INT8 model quantization to investigate privacy-aware and resource-efficient healthcare AI.
- Evaluated the federated anomaly detection framework, with the global model achieving **96.28% accuracy** and **0.978 AUROC** in the reported experimental evaluation.

Selected Research Projects

PromptPoisonLab

Adversarial Prompt Injection and LLM Security Benchmarking *2026*

Python, FAISS, SentenceTransformers, SQLite, Docker

- Developed an experimental framework for generating and analyzing adversarial prompts across **11 attack families**, supporting systematic investigation of instruction-following model vulnerabilities.
- Implemented evolutionary prompt mutation, embedding-based semantic deduplication, and structured experiment storage to support reproducible security evaluations and failure analysis.

MediTrack

Machine Learning-Based Healthcare Risk Assessment *2026*

Python, Django, Scikit-learn, Pandas, NumPy

- Developed a healthcare risk-assessment application using a Gradient Boosting classifier trained on the Pima Indians Diabetes dataset.
- Implemented a three-tier risk classification workflow and integrated model inference into a Django-based application.
- Evaluated predictive performance using class-specific F1 scores and macro-F1, achieving approximately **0.717 macro-F1** in the reported evaluation.

Technical Skills

Programming: Python, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, SciPy

Research Methods: Empirical Bayes, Statistical Bootstrapping, Federated Learning, Differential Privacy, Anomaly Detection, Model Evaluation, Transformers, Vision-Language Models

Research Tools: Git, GitHub, Docker, Linux, FAISS, Weights & Biases, Jupyter Notebook, Google Colab

Scientific Writing: LaTeX, BibTeX, IEEE/ACM manuscript preparation

Additional Information

TOEFL iBT: 100/120 (March 2026)